

Project

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1 Introduction

1.1 Social dynamic crisis

The landscape of global mental health has shifted drastically over the past years. According to the recent statistics from World Health Organization, there are over one billion people that are living with recognised mental health condition. This makes mental illness one of the leading drivers of global disability. The increase of people with these issues is strongly correlated with rising problems in social dynamics, most notably the psychological effects of COVID-19 pandemics.

The post pandemic era had major influence on traditional communication patterns. Extended periods of isolation have created strong reliance on digital first interactions, which caused widespread social anxiety. For many people navigating face to face interactions became stressful task, driving away people from physical support and pushing them to unhealthy coping strategies.

1.2 Economic crisis

However, human crisis is not the only problem when dealing with global mental health problem. Globally, there is a huge mismatch between the demand for the psychological support and economic resources that are allocated for it.

Massive macroeconomic crisis happens with low productivity, absenteeism, premature economic exit, which are often consequences of mental health struggles. In addition, despite the fact that globally around 8% of medical problems are related to psychological issues, only 2% of the overall health budget is allocated to mental health care, which is almost unchanged for almost a decade.

The systematic underfunding leaves majority of people without accessible help. In high income countries waiting list for clinical psychological evaluations routinely goes from six months to over a year. In low-to-middle income nations, the ratio of professionals to citizens drops significantly to as few as one psychiatrist per million people. Private therapy remains an luxury, leaving the vast majority of suffering populations with zero options for structured support.

1.3 The Gender Disparity in Medical Research

In the current system, there is evident neglect in women's mental health. Women are diagnosed with generalized anxiety disorder and major depressive disorders at twice the rate of men. Furthermore, younger women aged 16–24 are nearly three times as likely to experience an acute mental health issue as their male peers. Despite this clear epidemiological reality, medical and psychological research has

historically suffered from a profound male bias, creating a structural data deficit that leaves women at a distinct disadvantage when seeking help. This data gap is driven by three main historical and structural factors:

- Systematic trial exclusion that was present until mid 1990s, that happened because of the assumption that hormones would create noisy data.
- Data exploration bias: Because primary datasets were constructed using male subjects, the clinical baselines for diagnosis and pharmaceutical treatments used today are largely extrapolated from male physiology and psychology.
- Critical endocrine milestones unique to female biology, such as the postpartum phase, the premenstrual matrix, and perimenopause are drastically under researched, frequently leaving patients with answers that overlook the intersection of hormonal changes and neurological wellness.

1.4 Benefits of RAG inclusion

As a consequence of clinical bottlenecks and research gaps, individuals are increasingly using generative AI architectures for support. However, standard Large Language Models (LLMs) present severe deployment risks in sensitive domains, frequently generating hallucinations, or delivering unverified medical advice. To resolve this safely, Retrieval-Augmented Generation (RAG) introduces a strict architectural constraint. Instead of allowing an LLM to generate text solely from its internal weights, a RAG system forces the model to act as educated dashboard of a verified, external reference library. When a user submits a query, the system first passes it to a semantic vector database containing curated, open-source wellness data. The system retrieves the most relevant text chunks and uses them as a contextual ground for the Large Language Model. The LLM then generates a response that is strictly bound by these documents, completely avoiding medical prescriptions or the simulation of personal relationships.

General benefits of using RAG architecture are minimising the risk of hallucinations that is very common in LLMs, traceable source citations and cost-efficiency compared to fine-tuning, that can cost thousands of dollars.

When applied to women's mental health, a RAG framework allows a model to bypass the historical biases of mainstream medicine by building a curated, high-integrity knowledge base drawn directly from open-source lifestyle, nutritional, and psychoeducational literature. Crucially, because this system operates at the intersection of lifestyle adjustments, such as soundscapes, meditation preferences, somatic exercises, and endocrine-conscious nutritional adjustments, it sidesteps the clinical dangers of automated medical prescriptions. The architecture ensures complete user anonymity and provides free access to wellbeing protocols.

1.5 Establishing educational non-relational boundary

Fundamental part of this project is the complete rejection of the artificial or therapeutic personal relationships. Unlike conversational agents that attempt to simulate human empathy or act as artificial friends, this project is designed strictly for educational and knowledge-delivery purposes. The system treats the user not as a patient in a clinical relationship, but as a self-directed agent seeking high-integrity, evidence-based wellness data. By positioning the interface as an interactive reference dashboard rather than an intimate chat companion, the project eliminates the ethical hazard of misleading vulnerable users into believing they are conversing with an emotional entity. It focuses strictly on providing an objective opportunity for knowledge sharing and self-guided

improvement.

1.6 Existing wellbeing chatbots

Current leading paradigms in this digital mental health space are Woebot and Wysa. While both platforms gave successful access to psychological tools, there are gaps in operational and philosophical areas that RAG architecture aims to improve.

Historically, Woebot operates on a deterministic, rules based architecture rather than a fully generative one. While it utilizes Natural Language Processing (NLP) at the entry point, it uses this technology purely for input classification, mapping the user's words to specific intent categories. Once the intent is classified, the system routes the user down a pre scripted, human-authored conversational branching path rooted in Cognitive Behavioral Therapy (CBT). The primary limitation of these fixed scripts is that because every response originates from a static folder of pre-approved statements, the interaction can quickly feel repetitive. If a user's queries falls outside the pre-programmed tree, the illusion of understanding breaks down, leading to a mechanical and frustrating user experience that cannot adapt to highly individualized lifestyle queries.

Wysa was traditionally built using a similar hybrid model, combining structured AI guided paths with human-in-the-loop coaching. In recent iterations, the platform has begun integrating clinical Large Language Models to handle complex user inputs and draft weekly reflection summaries. However, because it operates as a commercial, clinical-first application, its main design goal remains relational. It explicitly seeks to establish a persistent digital therapeutic alliance and companion-oriented ongoing check-ins to build emotional compliance.

Architecture of RAG represents a significant evolutionary step forward from both rigid rules-based chatbots and unconstrained commercial conversational agents. It successfully unifies the strengths of natural language expression with absolute factual safety across three distinct technical are: First, regarding conversational flexibility, rules-based systems like traditional Woebot are limited to static branching paths, and unconstrained commercial systems present massive safety risks. The RAG architecture offers a dynamic yet bounded approach, where natural language generation remains fluid but completely pinned to external texts. Second, regarding the user relationship goal, while existing market players seek to build continuous companion-oriented bonds or digital alliances, this project implements a strictly educational model. It functions purely as an objective, high-integrity knowledge-sharing portal. Third, regarding risk management, RAG completely mitigates the high user friction and robotic repetition found in rules-based systems, while simultaneously eliminating the data hallucinations, prompt injections, or accidental medical diagnoses common to standard generative configurations. By transitioning away from rigid, pre-scripted buttons and moving toward a RAG architecture, this project provides users with a fluid, natural language interface that can answer highly specific, granular wellness questions. Simultaneously, by sourcing its knowledge directly from an explicit database of open-source literature on nutrition, somatic exercise, and mindfulness, it guarantees that the output remains entirely factual, traceably cited, and strictly non-medical.

2 Related Work

- Work that is connected to my thesis.
- Explanation of read research papers

3 Methodology

This chapter specifies the research design and implementation methodology for an educational chatbot that uses retrieval-augmented generation, a symptom-intervention graph, and nine coordinated software agents to answer questions concerning women’s mental health. The intended contribution is not an automated diagnostic or therapeutic system. It is an evidence-grounded educational system that retrieves traceable information, represents uncertainty, identifies situations that require professional or urgent care, and explains how mental, reproductive, sleep, nutritional, behavioral, and social factors may intersect. This distinction determines the architecture, data selection criteria, agent boundaries, evaluation measures, and safety controls described below.

The methodology follows a design-science and comparative experimental approach. A working artifact will be developed iteratively and evaluated against simpler baselines. The central hypothesis is therefore not that multiple agents are inherently superior, but that domain specialization, graph-supported evidence integration, and explicit safety gates can improve retrieval relevance, citation support, coverage of cross-domain symptom patterns, and safe uncertainty communication when compared with a single-agent RAG system. The design will be accepted only if these improvements are demonstrated empirically.

3.1 Research Questions and Scope

The primary research question is whether a graph-supported multi-agent RAG architecture produces more evidence-faithful, contextually complete, and safety-appropriate educational answers than a conventional single-agent RAG architecture for multi-symptom women’s mental health queries. The secondary questions examine which agents contribute measurable value, whether graph expansion improves retrieval of cross-domain evidence, whether confidence-based edge filtering reduces unsupported associations, and how well the system handles demographic and family-history context without making deterministic or discriminatory inferences.

The population scope includes adults seeking educational information about mood, anxiety, stress, fatigue, sleep, menstruation-related symptoms, pregnancy and postpartum mental health, perimenopause, nutrition, body-related concerns, and lifestyle factors. The system will be inclusive of people for whom reproductive or hormonal information is relevant and will not assume that every woman menstruates or that only women experience the conditions discussed. Age, reproductive stage, work pattern, caregiving responsibilities, cultural context, food and exercise access, social support, and self-reported family history may be used as optional context.

The chatbot will use non-diagnostic language such as ‘possible explanations include’, ‘this pattern can be associated with’, and ‘a clinician can help distinguish between’. It will not prescribe medication, advise medication discontinuation, provide individualized supplement dosing, or present a graph relation as proof of causation. The appropriate research outcomes are thus information-retrieval quality, grounding, safety behavior, calibration, clarity, and educational usefulness rather than clinical diagnostic accuracy.

3.2 Technical Foundations and Terminology

A large language model (LLM) is a neural network language model trained to predict and generate sequences of tokens. Most contemporary LLMs use the Transformer architecture, in which attention mechanisms model parallel relationships among tokens in context. A model’s parameters contain statistical regularities learned during training, but they do not constitute a verified or continuously updated medical database. Consequently, a fluent answer may be outdated, unsupported, or false. In this thesis, the LLM is treated as a controlled language generation component rather than central figure.

Retrieval-augmented generation (RAG) combines parametric memory in a language model with non-parametric memory stored in an external corpus. [1] introduced the influential RAG formulation for knowledge-intensive Natural Language Processing (NLP) tasks. In a practical application, documents are divided into chunks, encoded as vectors, stored with metadata, and retrieved in response to a user query. The retrieved passages are inserted into the generation prompt so that the model can produce an answer from explicit evidence. RAG supports source updating and provenance without retraining the generator; however, it does not guarantee factuality [2]. Retrieval can miss the best passage, irrelevant evidence can be selected, and a generator can still misuse correct evidence [3]. Health-care reviews therefore describe RAG as a reliability-enhancing method that still requires source validation, evaluation, and human governance [4].

Multi-agent RAG separates the workflow into role-bounded components. An agent is not necessarily a separate model. In this implementation, an agent is a software unit defined by a system prompt, an input and output schema, permitted tools, a domain-specific retrieval filter, a confidence rule, and a stop or escalation condition. Several agents may share one locally hosted instruction-tuned LLM while behaving differently because they receive different evidence and constraints. This design reduces hardware requirements and creates auditable intermediate outputs. Surveys of LLM-based multi-agent systems identify role specialization, communication, and coordination as principal design dimensions, but also note cost, error propagation, and evaluation challenges [5].

The symptom graph is a typed, attributed multigraph. Nodes represent symptoms, symptom clusters, conditions or educational topics, reproductive life stages, risk and protective factors, interventions, contraindications, evidence sources, and contextual variables. Edges represent qualified relations such as co-occurs-with, may-worsen, may-improve-with, is-associated-with, has-red-flag, contraindicated-for, and supported-by-source. The graph is not a diagnostic causal model. It is an evidence-navigation and ranking structure whose edges retain source identifiers and confidence values. GraphRAG research illustrates how graphs can support multi-hop and global evidence aggregation beyond isolated nearest-neighbor chunks [6].

Table 1: Terminology used in the proposed machine-learning framework.

Term	Operational meaning in this thesis
Token	A word fragment or symbol processed by the model. Token count determines context and generation length.
Context window	The maximum number of tokens the model can process in one request, including instructions, evidence, history, and output.
Embedding	A dense numeric vector representing semantic content so that related queries and passages can be compared.

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Table 1 continued

Term	Operational meaning in this thesis
Chunk	A bounded passage extracted from a source while retaining page, section, source, and license metadata.
Vector store	An index that stores embeddings and returns passages whose vectors are similar to a query vector.
Sparse retrieval	Term-based retrieval such as BM25, useful for exact symptoms, abbreviations, drug names, and uncommon terminology.
Dense retrieval	Embedding-based retrieval that captures semantic similarity even when query and source use different words.
Hybrid retrieval	Combination of sparse and dense rankings to improve coverage across exact and semantic matches.
Reranker	A slower model that scores query-passage pairs after initial retrieval and reorders the candidate evidence.
Grounding	Constraining an answer to claims supported by retrieved evidence and traceable citations.
Hallucination	Generated content that is unsupported, incorrect, or produced despite sounding plausible.
Prompt	Instructions and context supplied to the model, including role, evidence, output format, and safety rules.
Structured output	A validated object, normally JSON/Pydantic, with fixed fields rather than unrestricted prose.
Agent state	The shared record containing the query, route, evidence, graph paths, outputs, uncertainty, and safety flags.
Provenance	The source, page or timestamp, passage identifier, extraction method, and model version behind a claim or edge.
Confidence/calibration	A score and evaluation process indicating how often predictions at a stated confidence level are correct.
Precision and recall	Precision measures how many selected items are relevant; recall measures how many relevant items were found.
MRR and nDCG	Ranking metrics that reward placing relevant evidence near the top of the retrieved list.

3.3 Evidence-Based Justification for the Agent Domains

3.3.1 Mental health is multifactorial

The literature does not identify one universal or most common cause of mental health problems. The more defensible conclusion is that common mental disorders arise through interacting genetic, biological, psychological, behavioral, family, community, economic, cultural, and structural factors. The World Health Organization explicitly states that no single factor explains a mental health condition and describes risk and protection across individual and social environments (?). ?, in a systematic review of 289 reviews, organized social determinants into demographic, economic, neigh-

borhood, environmental-event, and social-cultural domains. NIMH similarly describes depression and generalized anxiety as resulting from combinations of genetic, biological, environmental, and psychological influences (??).

This multiagent model directly justifies specialist agents. It is methodologically preferable to asking one unconstrained model to compress every factor into a single explanation. The Mental Health Agent handles core affective, cognitive, trauma-related, and functional patterns. The Sleep Agent addresses a major bidirectional pathway. The Hormonal Health Agent captures reproductive timing and hormone sensitivity. The Nutrition and Body Agent covers dietary, somatic, eating, and supplement-related evidence; and the Lifestyle and Context Agent represents demographic and social determinants. The Intervention Agent then reasons over candidate actions only after the possible contributing domains and contraindications have been identified.

Family history is relevant but not deterministic. Research indicates that first-degree relatives of people with depressive or anxiety disorders have elevated average risk, but familial clustering combines genetic vulnerability with shared developmental and psychosocial conditions (?). The prototype will therefore treat self-reported family history as one contextual modifier. It will not infer a disorder, calculate a genetic risk score, or claim that a family history causes the user’s symptoms.

3.3.2 Common symptom and phrase families

The initial language taxonomy is derived from validated screening constructs and open medical guidance. The PHQ-9 validation study operationalizes depressed mood, loss of interest, sleep and appetite change, low energy, concentration difficulty, psychomotor change, self-critical thoughts, and self-harm ideation (?). The GAD-7 covers persistent worry, inability to control worry, nervousness, restlessness, difficulty relaxing, irritability, and fear that something bad may happen (?). NIMH guidance adds common somatic forms such as headaches, muscle tension, stomach symptoms, fatigue, and shortness of breath (?). Reproductive-health sources identify cycle-linked mood change, irritability, anxiety, appetite change, sleep disturbance, bloating, pain, and functional impairment in PMS/PMDD (?). Table 2 translates these clinical constructs into likely conversational phrase families. The phrases are examples for ontology design and testing, not diagnostic rules. A later corpus analysis should count their actual frequency in the collected question set, compare demographic and linguistic variation, and add misspellings and paraphrases only after privacy and ethics review.

Table 2: Conversational phrase families, normalized concepts, and agent routes.

Concept family	Example user language	Primary route
Low mood/anhedonia	'down', 'empty', 'numb', 'hopeless', 'nothing feels enjoyable'	Mental Health; Safety if severe
Anxiety/worry	'on edge', 'cannot stop worrying', 'my mind will not switch off', 'something bad will happen'	Mental Health; Sleep; Context
Energy/fatigue	'drained', 'tired all the time', 'no energy', 'heavy', 'burned out'	Sleep; Mental Health; Hormonal; Nutrition
Cognition	'brain fog', 'cannot focus', 'forgetful', 'cannot decide', 'slowed down'	Mental Health plus relevant specialists

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Table 2 continued

Concept family	Example user language	Primary route
Sleep	'cannot fall asleep', 'wake during the night', 'wake too early', 'sleep too much'	Sleep; Mental Health; Hormonal
Irritability/overload	'snapping at everyone', 'mood swings', 'overwhelmed', 'out of control'	Mental Health; Hormonal; Context
Appetite/body	'craving', 'overeating', 'no appetite', 'bloating', 'weight change', 'body distress'	Nutrition and Body; Hormonal; Mental Health
Cycle timing	'worse before my period', 'better after it starts', 'monthly crash'	Hormonal; Mental Health; Sleep
Reproductive stage	'postpartum and not coping', 'perimenopause', 'hot flashes and low mood'	Hormonal; Mental Health; Safety
Somatic distress	'tight chest', 'headaches', 'muscle tension', 'stomach pain', 'heart racing'	Mental Health; Safety; Body
Functional impairment	'cannot work or study', 'cannot care for myself', 'avoiding everyone'	Mental Health; Safety; all routed domains
Eating-risk language	'not eating', 'bingeing', 'purging', 'terrified of weight gain'	Safety; Nutrition and Body; Mental Health
Crisis/euphemism	'want to disappear', 'better off dead', 'hurt myself', 'not safe at home'	Immediate Safety route
Social determinants	'money stress', 'caregiving', 'shift work', 'violence', 'discrimination', 'no support'	Lifestyle and Context; Safety

3.3.3 Cross-domain evidence supporting specialization

Sleep disturbance warrants a dedicated agent because its relationship with depression and anxiety is frequently bidirectional. A systematic review by ? found evidence for reciprocal prospective relationships among sleep disturbance, anxiety, and depression, while later reviews continue to treat insomnia as both a symptom and a risk pathway. A Sleep Agent can therefore distinguish sleep-onset difficulty, sleep maintenance, early waking, hypersomnia, circadian disruption, shift work, and daytime impairment, while referring mood and safety implications to the relevant agents. Reproductive timing warrants a Hormonal Health Agent because menstrual, perinatal, and menopausal transitions can change symptom interpretation and risk. PMDD includes affective, cognitive, behavioral, sleep, appetite, and physical symptoms whose cyclical timing and functional impairment are central to recognition (??). A recent comprehensive review concludes that women's reproductive mental health has complex biological, psychological, social, cultural, and political foundations and requires an interdisciplinary approach (?). The agent must therefore avoid simplistic statements such as 'hormonal imbalance causes depression' and instead ask about timing, recurrence, impairment, reproductive stage, medication, and professional assessment.

Nutrition, body experience, and physical activity warrant bounded specialist treatment because evidence strength differs by intervention. A meta-analysis of randomized trials found that dietary improvement produced a small reduction in depressive symptoms, with no statistically significant pooled effect for anxiety, and emphasized heterogeneity and the need for further study (?). A large

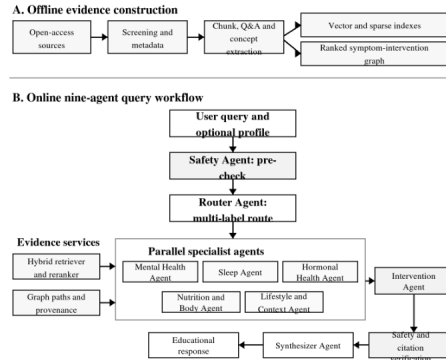


Figure 1: Proposed architecture for the nine-agent graph-supported RAG system. The offline pipeline produces a vector/sparse index and a provenance-rich symptom-intervention graph. During online use, the Safety and Router Agents precede parallel specialist retrieval; the Intervention and Synthesizer Agents operate only on structured, cited outputs, followed by safety and citation verification.

network meta-analysis found several exercise forms effective for depression, while the authors also reported variation in certainty and study quality (?). Music therapy and mindfulness have evidence in selected contexts, but effect estimates, intervention definitions, and certainty vary (??). These findings support an Intervention Agent that ranks evidence and distinguishes low-risk education from clinician-led treatment rather than listing every plausible activity as equally established.

3.4 Proposed Nine-Agent Architecture

The proposed system contains nine logical agents: Router, Safety, Mental Health, Sleep, Hormonal Health, Nutrition and Body, Lifestyle and Context, Intervention, and Synthesizer. The Safety Agent is reused at several points but counts as one logical agent. The five domain specialists may run in parallel after routing, while safety, intervention filtering, synthesis, and final verification remain ordered. Figure ?? separates the offline evidence-construction pipeline from the online query workflow.

Table 3: Operational responsibilities of the nine logical agents.

Agent	Bounded responsibility	Structured output
Router	Multi-label domain classification and query rewriting	Agent list, rewritten queries, route confidence
Safety	Crisis, self-harm, eating-risk, abuse, urgent medical, medication and supplement safeguards	Risk level, escalation, allowed answer scope
Mental Health	Mood, anxiety, trauma/stress, cognition, somatic expression, comorbidity and impairment	Mental-health concepts, evidence, uncertainty

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Table 3 continued

Agent	Bounded responsibility	Structured output
Sleep	Insomnia, hypersomnia, circadian timing, sleep deprivation, fatigue and sleep-related red flags	Sleep pattern and evidence-supported education
Hormonal Health	Cycle timing, PMS/PMDD, pregnancy/postpartum, perimenopause and reproductive context	Timing hypotheses, tracking questions, referral cues
Nutrition and Body	Diet quality, appetite, caffeine, possible deficiencies, body image, eating risk and supplement cautions	Evidence-graded nutrition/body information
Lifestyle and Context	Age, work, caregiving, activity, stress, culture, access, support, substance use and family history	User-provided context modifiers and access-aware options
Intervention	Links supported symptom pathways to low-risk education or clinician-discussion options	Ranked actions, evidence grade, contraindications
Synthesizer	Reconciles agent outputs and writes one concise educational answer	Cited answer, uncertainty, red flags and follow-up

3.4.1 Router Agent

The Router Agent performs multi-label classification because one query may require several specialists. It receives the raw query, a summary of relevant conversation history, and optional user-provided context. It returns domain labels, a query reformulation for each domain, confidence values, and a list of follow-up fields that are necessary before safe answering. Routing should first use a lightweight supervised classifier such as DeBERTa-v3, with an LLM structured-output fallback for ambiguous cases. A rule layer ensures that crisis phrases always activate Safety and that explicit cycle, pregnancy, postpartum, or perimenopause terms activate Hormonal Health. Routing accuracy will be evaluated against manually labelled multi-domain questions.

3.4.2 Safety Agent

The Safety Agent is a gate rather than an ordinary specialist. It runs on the raw query, on candidate interventions, and on the draft answer. The first pass identifies imminent self-harm or suicide risk, inability to remain safe, abuse or coercion, severe eating-disorder behavior, acute confusion, pregnancy or postpartum emergencies, severe bleeding, chest pain, breathing difficulty, medication discontinuation requests, and dangerous supplement combinations. High-risk input bypasses normal educational synthesis and triggers supportive, localized crisis or emergency guidance. The final pass checks that no diagnosis, individualized prescription, unsafe dose, unsupported reassurance, or delay-of-care message has been introduced. A hybrid of deterministic rules, a calibrated classifier, and constrained LLM review is preferable because euphemistic crisis language may evade keywords while an LLM alone may be inconsistent.

3.4.3 Mental Health Agent

The ninth and newly added Mental Health Agent owns the core mental-health interpretation that was previously distributed implicitly across other agents. It covers depressive and anxiety symptom clusters, trauma- and stress-related language, rumination, irritability, anhedonia, panic-like symptoms, concentration difficulty, somatic expressions, social withdrawal, functional impairment, co-occurrence, and differential educational questions. It does not diagnose major depressive disorder, generalized anxiety disorder, bipolar disorder, PTSD, or another condition. Instead, it identifies symptom dimensions, duration and impairment questions, evidence-supported possibilities, and red flags. This dedicated role prevents the Lifestyle Agent from becoming a substitute mental-health clinician and allows Sleep or Hormonal outputs to be interpreted alongside, rather than in place of, primary mood and anxiety evidence.

3.4.4 Sleep Agent

The Sleep Agent retrieves only from the sleep and circadian subcorpus plus cross-domain graph neighbors. It classifies sleep-onset insomnia, sleep-maintenance insomnia, early waking, hypersomnia, irregular schedule, shift-work disruption, nightmares, suspected sleep-disordered breathing, and daytime fatigue. The output distinguishes educational sleep-hygiene information from signs that merit clinical evaluation. Because sleep and mood influence one another, the agent must never conclude that sleep is the sole cause of a user’s distress; it reports a bidirectional pathway and hands mood severity or self-harm indicators to Mental Health and Safety.

3.4.5 Hormonal Health Agent

The Hormonal Health Agent evaluates temporal patterns around the menstrual cycle and reproductive life stages. Its concepts include PMS, PMDD, premenstrual exacerbation of another condition, dysmenorrhea, pregnancy, postpartum, lactation, perimenopause, menopause, contraception, thyroid-related educational differentials, and symptom tracking. A key function is to request timing information: when symptoms begin, whether they improve after menstruation, whether they recur across cycles, and how much they impair daily functioning. The agent communicates that sensitivity to hormonal fluctuation varies across individuals and that similar symptoms can arise from non-hormonal mental or physical conditions.

3.4.6 Nutrition and Body Agent

The Nutrition and Body Agent addresses dietary patterns, appetite and cravings, caffeine and alcohol, low energy availability, possible deficiency-related educational topics, gastrointestinal overlap, body image distress, exercise, and eating-disorder warning signs. It also retrieves contraindication evidence for herbs and supplements. Supplements are treated as safety-sensitive entities: an apparently positive efficacy edge is insufficient without evidence quality, medication interaction, pregnancy or lactation status, allergy, dose uncertainty, and clinician-discussion requirements. The agent must not infer a deficiency from fatigue or recommend laboratory testing as a diagnosis; it may explain that a clinician can investigate physical conditions when symptoms are persistent or severe.

3.4.7 Lifestyle and Context Agent

The Lifestyle and Context Agent interprets only information voluntarily provided by the user. Its purpose is not to blame lifestyle for illness, but to identify constraints and social determinants that change the feasibility or relevance of education. Variables include age, work and shift schedule, caregiving load, physical activity, substance use, housing and food access, culture, discrimination, social support, relationship safety, and self-reported family history. It can replace an inaccessible suggestion with a lower-cost or culturally compatible alternative and can ask a minimal follow-up question when context materially changes safety. Genetic claims are excluded from the initial system; family history remains a probabilistic context feature and is not passed to the generator as a diagnosis.

3.4.8 Intervention Agent

The Intervention Agent operates after domain evidence has been collected. It maps supported symptom pathways to candidate actions, then applies evidence thresholds, user constraints, contraindications, and the Safety Agent’s scope. Outputs are grouped into: low-risk educational actions, topics to discuss with a qualified professional, and urgent or crisis actions. Exercise, sleep scheduling, symptom tracking, mindfulness, music-based activities, diet quality, and social support may be discussed when evidence and context permit; psychotherapy and medication are described as professional treatment categories; supplements and herbs receive conservative handling. The agent must return an empty list or an abstention when the evidence is insufficient.

3.4.9 Synthesizer Agent

The Synthesizer Agent receives structured outputs rather than unrestricted agent conversations. It ranks the most plausible educational explanations, states what is uncertain, identifies overlap and contradiction, and cites the underlying passages. The final answer should acknowledge the concern, explain two or three evidence-supported possibilities, identify red flags, present proportionate next steps, and ask only necessary follow-up questions. Every factual health claim must map to at least one retrieved passage identifier. If a claim lacks support, it is removed by a citation-verification function before the final Safety pass.

3.5 How the Agents Are Built in Code

The implementation begins with shared schemas and one reusable agent runner, not nine independent chatbots. Each AgentSpec contains a name, system prompt, metadata filters, allowed tools, output schema, minimum evidence requirement, and confidence threshold. The generator is loaded once and reused. The specialist runner performs domain-filtered retrieval, reranking, graph expansion, prompt construction, structured generation, and validation. This architecture makes differences among agents explicit and supports ablation experiments by disabling one specification at a time.

Listing 1: Simplified agent contract and reusable domain runner.

```
class AgentSpec(BaseModel):
    name: str
    domains: list[str]
    system_prompt: str
```

```

tools: list[str]
min_evidence: int = 2

class AgentOutput(BaseModel):
    concepts: list[str]
    evidence_ids: list[str]
    graph_paths: list[str]
    confidence: float
    uncertainty: list[str]
    safety_flags: list[str]

def run_domain_agent(spec: AgentSpec, state: RAGState) -> AgentOutput:
    candidates = hybrid_retrieve(state.user_query, filters=spec.domains)
    evidence = rerank(state.user_query, candidates)[:TOP_N]
    graph_context = graph_expand(extract_query_concepts(state.user_query))
    raw = generator.with_structured_output(AgentOutput).invoke(
        build_prompt(spec, state, evidence, graph_context)
    )
    return validate_output(raw, allowed_evidence=evidence)

```

Pydantic schemas enforce field types and make missing evidence visible. Each output contains evidence identifiers rather than copied paragraphs, limiting prompt growth and allowing later citation verification. The validator rejects outputs with unknown evidence IDs, out-of-range confidence, prohibited diagnostic wording, or intervention claims outside the agent’s permissions. The same design can be implemented with a custom Python orchestrator or a state-graph framework such as LangGraph. LangGraph is useful when conditional routes, retries, and safety branches must be visible; a custom orchestrator remains a valid baseline and reduces framework dependence.

3.5.1 Shared state and orchestration

All nodes read and update one typed RAGState. Required fields are `user_query`, optional `user_profile`, `safety_flags`, `selected_routes`, `evidence_by_agent`, `graph_paths`, `agent_outputs`, `draft_answer`, `verified_citations`, and `final_answer`. Safety pre-check and routing are sequential. The selected specialist agents can execute concurrently because each writes to a separate output key. The Intervention Agent waits for all specialists, and the Synthesizer waits for intervention filtering. Retries are permitted only for invalid structured output or temporary model failure; they are not used to pressure an agent into producing an answer when evidence is absent.

Listing 2: Online orchestration in implementation-oriented pseudocode.

```

def answer_query(query: str, profile: UserProfile | None) -> FinalAnswer:
    state = RAGState(user_query=query, user_profile=profile)
    state.safety_flags = safety_agent.precheck(state)
    if state.safety_flags.requires_immediate_escalation:
        return safety_agent.crisis_response(state)

    state.selected_routes = router_agent.route(state)
    state.agent_outputs = run_selected_specialists_in_parallel(state)

```

```

state.interventions = intervention_agent.rank_and_filter(state)
state.draft_answer = synthesizer_agent.compose(state)
state.verified_citations = verify_claim_support(state.draft_answer)
return safety_agent.postcheck_and_release(state)

```

3.6 Knowledge Base Construction

3.6.1 Source selection and evidence hierarchy

The corpus will include open-access systematic reviews and meta-analyses, peer-reviewed primary studies, clinical or public-health guidelines, official health-agency materials, open textbooks, and licensed podcast transcripts. Every source will be catalogued with title, authors, year, publication type, DOI or URL, license, peer-review status, population, domain, and evidence tier. The preferred hierarchy is: guidelines and systematic reviews; randomized or strong longitudinal studies; official patient guidance; narrative reviews and textbooks; then expert podcasts or general educational transcripts. The last category may provide accessible phrasing and question forms but must not independently support a medical claim or intervention edge.

Inclusion requires relevance to the declared scope, a traceable publisher or institutional source, usable licensing or quotation rights, sufficient methodological description, and extractable provenance. Exclusion criteria include anonymous wellness claims, commercial supplement pages, sources without verifiable references, diagnosis-promoting self-tests outside validated instruments, and content that encourages medication changes without clinical supervision. Conflicting high-quality sources are retained and marked rather than silently removed, because contradiction is a meaningful uncertainty signal.

3.6.2 Cleaning, chunking, and metadata

PDFs, web pages, books, and transcripts are converted into normalized document objects while preserving headings, page numbers, tables, speaker labels, and timestamps. Repeated headers, footers, navigation text, advertisements, and reference-only fragments are removed. Initial chunking will use semantically coherent units of approximately 250-500 tokens with 50-100 token overlap, followed by experiments with section-aware and sentence-window chunking. Character counts alone are avoided because a 300-character chunk is often too short to preserve a medical claim and its qualification. Each chunk inherits source and domain metadata and receives a stable identifier.

Metadata fields include `source_id`, `chunk_id`, `title`, `author`, `year`, `page_or_timestamp`, `section`, `source_type`, `domain_labels`, `population`, `life_stage`, `evidence_tier`, `peer_reviewed`, `license`, `safety_tags`, and `ingestion_version`. These fields permit domain filtering, evidence weighting, reproducible re-indexing, and auditing of which sources influence an answer.

3.6.3 Q&A and graph-candidate extraction

Question-answer pairs are extracted for evaluation and possible later fine-tuning. A structured extraction prompt returns a natural-language question, a concise answer, exact supporting passage spans, domain labels, concept identifiers, life-stage tags, safety tags, and an answer type. Pairs without a supporting span are discarded. Automated entailment or reranker checks flag weak pairs, and a stratified sample is manually reviewed. The initial use of these pairs is evaluation; fine-tuning is deferred until the baseline reveals a repeatable failure that retrieval and prompting cannot solve.

Candidate graph triples are extracted from the same chunks. Each triple contains `source_node`, `relation`, `target_node`, `evidence_ids`, `extraction_method`, `source_quality`, and `extraction_confidence`. Concepts are normalized to a controlled vocabulary while retaining user synonyms. Duplicate edges are merged only when relation direction and meaning match. Statements of association are not converted to causal edges, and intervention edges remain separate from condition edges to prevent a co-occurrence relation from being interpreted as a recommendation.

3.7 Retrieval, Reranking, and Confidence

3.7.1 Hybrid retrieval

The retriever combines sparse BM25 search with dense embedding search. Sparse retrieval is valuable for exact entities such as PMDD, SSRI, chasteberry, ferritin, or a quoted symptom phrase. Dense retrieval is valuable when the user’s wording differs from the literature. Candidate rankings are combined with reciprocal rank fusion, after which a cross-encoder scores each query-passage pair. Dense Passage Retrieval and Sentence-BERT demonstrate the effectiveness of learned dense representations, while retrieve-then-rerank architectures improve precision by applying a more expensive pairwise model only to a small candidate set (??).

Retrieval is domain filtered but not completely isolated. Each specialist first searches its primary corpus and then receives a limited cross-domain expansion when graph paths or query concepts indicate overlap. For example, a query about exhaustion before menstruation may retrieve from Hormonal Health, Sleep, Mental Health, and Nutrition and Body. The Router does not remove evidence; it controls the search budget. If routing confidence is low, the system activates a broader retrieval fallback rather than prematurely excluding a domain.

3.7.2 Graph-edge ranking

Automatic relation extraction is noisy, so the graph stores candidate and accepted edges separately. A transparent initial score combines source quality, semantic support from the reranker, cross-source agreement, extraction confidence, agent consensus, and guideline or recency relevance, then subtracts contradiction and safety penalties. The proposed score is shown below; weights are hypotheses to be tuned on a manually labelled validation set rather than fixed truths.

$$\begin{aligned}
 s_{\text{edge}} = & 0.25q_{\text{source}} + 0.25s_{\text{semantic}} + 0.15a_{\text{cross-source}} \\
 & + 0.10c_{\text{extraction}} + 0.10c_{\text{agent}} + 0.10p_{\text{relation}} \\
 & + 0.05r_{\text{guideline/recency}} - 0.20p_{\text{contradiction}} - 0.15p_{\text{safety}}.
 \end{aligned} \tag{1}$$

Edges above a high threshold may be used in answers if their cited passages support the exact relation. Medium-score edges can prompt a follow-up question or be stated with explicit uncertainty. Low-score edges remain hidden from users. Safety-sensitive edges involving self-harm, medication changes, supplements, pregnancy, lactation, eating disorders, or urgent symptoms require Safety approval regardless of score. Calibration will compare predicted confidence with human edge correctness; isotonic regression or Platt scaling can be applied if raw scores are overconfident.

3.8 How the Graph Is Included in the Code

The first prototype will use NetworkX MultiDiGraph because it is lightweight, inspectable, and sufficient for thesis-scale experiments. Neo4j becomes appropriate if the graph grows large, multi-hop

queries become a primary interface, or concurrent persistence is required. Graph code is separated into `schema.py`, `extract.py`, `rank.py`, `store.py`, and `query.py`. This separation allows the extraction model, score formula, or storage engine to change without altering the agents.

Listing 3: Minimal provenance-rich graph insertion.

```
import networkx as nx

graph = nx.MultiDiGraph()
graph.add_node("symptom:insomnia", type="Symptom", label="Insomnia")
graph.add_node("symptom:anxiety", type="Symptom", label="Anxiety")
graph.add_edge(
    "symptom:insomnia",
    "symptom:anxiety",
    relation="may_worsen",
    evidence_ids=["src_014:p6:c03", "src_091:p2:c08"],
    score=0.84,
    status="accepted",
)

def graph_expand(concepts, max_hops=2, min_score=0.75):
    return ranked_paths(graph, concepts, max_hops, min_score)
```

At query time, a concept extractor maps phrases such as 'I cannot sleep and feel on edge before my period' to normalized nodes for insomnia, anxiety, and premenstrual timing. Graph expansion retrieves only one- or two-hop paths above the domain-specific threshold. The path is then converted into a compact statement with evidence IDs; the supporting chunks, not the graph label alone, are passed to the generator. This rule prevents the graph from becoming an unverified source of truth.

The Intervention Agent uses restricted traversals such as `symptom → may-improve-with → intervention` and `intervention → contraindicated-for → context`. A proposed intervention is rejected if a contraindication path is supported, if the evidence does not match the population, or if the Safety Agent requires clinician supervision. Graph provenance enables an examiner to trace any visible connection back to a passage and extraction version.

3.9 Model and Software Selection

The architecture requires different model classes. The generator is an instruction-tuned causal language model that writes structured and final text. The embedding model creates retrieval vectors. A cross-encoder reranks passages. Encoder models classify domains, safety signals, concepts, and relations. MentalBERT and DeBERTa are therefore not substitutes for the answer-generating LLM; they are candidate encoders for narrower tasks. This division improves efficiency and makes each component independently testable.

Table 4: Recommended free/open model stack and methodological role.

Model	Role	Rationale and constraint
Qwen2.5-7B-Instruct	Primary local generator	Apache 2.0; strong instruction following and JSON output; practical 4-bit baseline.
Qwen3-8B	Secondary generator experiment	Apache 2.0; stronger agent/tool orientation; thinking mode adds latency and evaluation complexity.
Phi-4-mini-instruct	Low-resource fallback	MIT; smaller and suited to constrained hardware; depends more heavily on retrieval quality.
DeBERTa-v3-base	Router/safety/relation classifier	MIT; strong general NLU encoder; requires labelled task data and calibration.
MentalBERT	Research comparison encoder	Mental-health-domain language model; gated and CC BY-NC 4.0; social-media domain shift and non-diagnostic limitation.
SentenceTransformer + cross-encoder	Retriever and reranker	Efficient retrieve-then-rerank stack; model choice must be evaluated on the thesis corpus.

Qwen2.5-7B-Instruct is the recommended first generator because its official model card describes Apache 2.0 licensing, instruction following, structured-output capability, and long-context support (?). A 4-bit quantized deployment can fit common research GPUs, although actual memory depends on context, cache, and runtime. Qwen3-8B is a valuable second experiment for tool use and reasoning (?), while Phi-4-mini-instruct is the pragmatic fallback when memory or latency is the limiting factor. 'Free to download' does not remove compute cost or licensing obligations; the thesis will record model version, revision hash, quantization, runtime, and license.

MentalBERT was pretrained on mental-health-related Reddit posts and its model card explicitly limits its role to non-clinical use and states that predictions are not psychiatric diagnoses (?). It is useful as an experimental feature encoder or classifier but may inherit platform-specific language and demographic bias. DeBERTa-v3 has an MIT license and is a cleaner default for supervised routing and safety classification (?). MentalBERT should be compared against it only after a labelled dataset is available.

3.10 Coding Pipeline and Implementation Order

3.10.1 Repository and environment

Coding should begin with the data and evaluation contracts, then a single-agent baseline. Implementing all nine agents before validating retrieval would make errors difficult to localize. A Python 3.10 or later environment can use PyTorch, Transformers, Accelerate and bitsandbytes for local generation; SentenceTransformers, rank-bm25 and FAISS or Chroma for retrieval; Pydantic for schemas; NetworkX for the initial graph; LangGraph or a custom orchestrator for state transitions; FastAPI for an API; and pytest, scikit-learn and a RAG evaluation toolkit for testing. Exact versions will be locked in requirements.txt or pyproject.toml.

Listing 4: Suggested repository structure.

```
thesis-rag-chatbot/  
  data/raw/                # licensed PDFs, pages, transcripts  
  data/processed/          # normalized documents and chunks  
  data/qa_pairs/           # generated, validated and reviewed Q&A  
  app/schemas.py          # shared typed inputs and outputs  
  app/agents/              # nine AgentSpec definitions and prompts  
  app/retrieval/           # BM25, embeddings, fusion and reranking  
  app/graph/               # schema, extraction, ranking, store, query  
  app/safety/              # rules, classifier and response policies  
  app/orchestration/       # shared state and workflow graph  
  app/evaluation/          # retrieval, graph, answer and safety metrics  
  app/api/                 # FastAPI endpoints or research interface  
  scripts/build_index.py   # reproducible offline indexing  
  scripts/build_graph.py   # candidate extraction and edge acceptance  
  tests/                   # unit, integration and regression tests
```

3.10.2 Step-by-step implementation

1. Define Pydantic schemas for Source, EvidenceChunk, UserProfile, AgentSpec, AgentOutput, GraphNode, GraphEdge, SafetyDecision, RAGState, and FinalAnswer. Write validation tests before loading an LLM.
2. Create the source catalogue and ingestion functions. Preserve provenance and license metadata, then manually inspect a sample of cleaned and chunked documents.
3. Build the sparse and dense indexes. Create a small gold set of questions with relevant passage IDs and report Recall@5, Recall@10, MRR, and nDCG before generation is introduced.
4. Add a cross-encoder reranker and compare hybrid retrieval with dense-only and sparse-only baselines. Select top-k and top-n values on validation data rather than intuition.
5. Load one local generator and implement a single-agent RAG answer with evidence IDs, citations, abstention, and a final claim-support check. This becomes Baseline A.
6. Implement Safety and Router nodes. Label a multi-domain and crisis test set, train or fine-tune DeBERTa-v3, calibrate thresholds, and retain deterministic override rules.
7. Add the five specialist AgentSpec configurations one at a time. Each addition must pass retrieval, schema, citation, and safety tests before parallel execution is enabled.
8. Build the concept normalizer and candidate graph extractor. Manually label graph edges, tune the confidence threshold, and expose only accepted edges with provenance.
9. Add the Intervention and Synthesizer Agents. Enforce contraindication traversal, evidence grading, citation verification, and final Safety release.
10. Expose the workflow through FastAPI or a restricted research interface, log non-identifying run metadata, freeze an evaluation version, and conduct baseline, ablation, usability, and red-team experiments.

3.10.3 Model loading

Hugging Face Transformers downloads model weights through `from_pretrained` and caches them locally. Qwen2.5 and Qwen3 can be loaded directly under their stated licenses; MentalBERT requires authentication and acceptance of gating conditions. Four-bit loading reduces memory but may alter output quality, so quantized and full-precision or higher-precision results should not be compared without recording the configuration. The following listing is a starting point rather than a production deployment recipe.

Listing 5: Local 4-bit generator loading.

```
from transformers import AutoModelForCausalLM, AutoTokenizer, BitsAndBytesConfig

MODEL_ID = "Qwen/Qwen2.5-7B-Instruct"
quant = BitsAndBytesConfig(load_in_4bit=True)
tokenizer = AutoTokenizer.from_pretrained(MODEL_ID)
model = AutoModelForCausalLM.from_pretrained(
    MODEL_ID,
    device_map="auto",
    torch_dtype="auto",
    quantization_config=quant,
)
model.eval()
```

3.11 Safety, Privacy, and Ethical Governance

WHO guidance requires autonomy, well-being, transparency, accountability, inclusiveness, and sustainability to remain central in health AI design (?). For this project, those principles become concrete engineering requirements: the system discloses its educational scope; allows users to omit profile fields; shows citations and uncertainty; records model and source provenance; tests performance across demographic and phrasing variation; and uses a clear escalation policy. RAG may improve traceability but can also retrieve biased or unsafe material, so curation and final verification remain necessary.

The prototype should avoid storing identifiable health conversations unless ethics approval, informed consent, access control, retention limits, and data-protection procedures are established. Development logs should use synthetic or de-identified scenarios and record component scores rather than raw sensitive text where possible. Prompt-injection defenses must treat retrieved documents as untrusted data: retrieved text cannot alter system instructions, request secrets, or redefine safety rules. Source ingestion should scan for malicious instructions and preserve a strict separation between evidence content and executable policy.

Safety evaluation must be localized. A global system cannot rely only on the United States 988 number. The interface should request or infer location only with consent and when necessary for crisis resources, otherwise instruct the user to contact local emergency services or a local crisis line. The educational non-relational boundary should also be retained: the chatbot can communicate respectfully and supportively without claiming to be a therapist, friend, conscious companion, or substitute for human care.

3.12 Evaluation Design

The evaluation will compare four systems under the same generator and corpus: (A) parametric LLM without retrieval; (B) single-agent RAG; (C) multi-agent RAG without graph expansion; and (D) the complete multi-agent graph RAG system. This staged comparison isolates the effects of retrieval, specialization, and graph support. Additional ablations remove the reranker, individual specialists, confidence filtering, and final citation verification. Multiple random seeds or repeated generations are required because decoding is stochastic.

The test set will combine reviewed source-derived Q&A pairs with manually written scenarios. Scenarios will include single-domain questions, ambiguous multi-symptom questions, cycle-linked and reproductive-stage questions, demographic and access constraints, family-history questions, supplement and herb questions, contradictory evidence, insufficient-evidence cases, adversarial prompt injections, eating-risk language, and direct or euphemistic crisis messages. A held-out test set will not be used to choose thresholds or prompts.

Table 5: Evaluation dimensions and research questions.

Dimension	Measures	Interpretation
Retrieval	Recall@k, MRR, nDCG, reranker gain, domain coverage	Were the passages needed for a correct answer retrieved and ranked early?
Graph	Edge precision/recall, duplicate rate, contradiction rate, calibration	Are accepted relations supported, typed correctly, and appropriately confident?
Routing	Micro/macro F1, exact multi-label match, safety recall	Were all necessary specialists selected without excessive activation?
Generation	Claim support, citation correctness, completeness, abstention	Does every material claim follow from cited evidence?
Safety	Crisis sensitivity, false escalation, prohibited-advice rate	Does the system escalate genuine risk and avoid unsafe medical direction?
User quality	Clarity, usefulness, non-stigma, cognitive load	Can users understand possibilities, uncertainty, and appropriate next steps?
Efficiency	Latency, GPU memory, tokens, retrieved chunks, agent count	Is any quality gain proportionate to computational cost?

At least two trained reviewers should independently rate a stratified sample using a predefined rubric, with disagreements resolved by discussion and inter-rater agreement reported. Clinician or qualified mental-health review is strongly preferable for safety-sensitive scenarios. Automated LLM-as-judge scores may support error analysis but must not be the only evaluation because the judge can share the generator’s biases. Statistical analysis should report confidence intervals and paired comparisons, while qualitative error analysis should categorize retrieval miss, routing miss, graph error, unsupported synthesis, over-escalation, under-escalation, and unclear language.

3.13 Expected Benefits and Limitations

The expected benefit of the proposed architecture is controlled decomposition. Each specialist searches a narrower evidence space; the Router allocates computation; the graph exposes cross-domain paths; the Intervention Agent applies evidence and contraindication rules; and the Synthesizer produces one answer with traceable claims. Structured intermediate outputs support debugging, ablation, and supervisor inspection. The design also permits individual components to be updated without retraining the full generator and can run mostly with free or open-weight software. Evidence from multi-agent clinical simulations suggests that agent discussion can improve performance in selected tasks, but those findings do not establish safety or generalizability for a public mental-health chatbot (?).

The limitations are substantial. More agents increase latency, token use, and the number of failure points. Agents that share one model may reproduce the same bias rather than provide independent judgment. Routing can omit a relevant domain; retrieval can surface low-quality evidence; automatic graph extraction can convert correlation into causation; and confidence scores can be poorly calibrated. Open-access literature may underrepresent some populations and languages, while podcast-derived phrasing may reflect speaker or platform bias. These limitations are reasons for baseline comparison, provenance, conservative thresholds, human review, and abstention rather than reasons to claim that the architecture eliminates hallucination.

Accordingly, the contribution of the thesis is an evaluated educational information architecture, not a claim of clinical equivalence. Success would be demonstrated by measurable improvements in evidence retrieval, claim support, cross-domain coverage, and safety behavior, accompanied by transparent reporting of cases in which the simpler baseline performs equally well or better.

3.14 References

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4 Results and Evaluation

- Analysis of results
- Comparison of different approaches
- Limitations and extensions of current work

5 Conclusion